FMCG Warehouse: Optimizing Amazon's Distribution Efficiency

Background

You are a data analyst at Amazon, which has an extensive network of warehouses across various regions. The company is facing challenges in ensuring efficient warehouse operations, resulting in frequent stockouts, delays in deliveries, and increased operational costs. Management has tasked you with analyzing warehouse data to identify key issues and provide actionable insights to enhance warehouse efficiency.

Problem Statement

Problem Statement: Amazon is experiencing inefficiencies in its warehouse operations. These inefficiencies affect the supply chain, leading to stockouts, delivery delays, and increased costs. Analyze the warehouse data to investigate operational patterns and propose evidence-based areas for review.

Why is it important to solve? Addressing warehouse inefficiencies can support customer satisfaction, reduce avoidable operational costs, and improve timely delivery. Efficient warehouse operations help maintain a competitive position in retail.

Stakeholders

Types of stakeholders:

- **Internal**: Warehouse Managers, Supply Chain Analysts, Operations Team, Senior Management.
- **External**: Retail partners, Customers, Logistics Providers.

Problem Definition:

Identify useful warehouse performance indicators and areas that need improvement. Analyze available fields to understand which operational characteristics are associated with reported refill, transport, storage, inspection, and breakdown activity.

Data Requirements:

- Warehouse capacity and Utilization
- Refill requests and transport issues
- Competitor presence
- Number of retail shops served
- Product weight and storage requirements
- Electric supply and distance from the hub
- Worker numbers and warehouse establishment year
- Issues reported, government certifications, and inspections

Metric Development: Develop metrics that the workbook can support, including average refill requests, transport issues, storage issues, breakdowns, and government checks per warehouse or group. Stockout rate, delivery-delay frequency, and operational cost per unit are useful business concepts but cannot be calculated from the listed fields without additional data.

Insights and Actions: Compare warehouse groups to locate patterns that merit investigation. Potential follow-up actions include reviewing refill schedules, storage conditions, maintenance practices, staffing, and transport routes. Recommendations should be proportionate to the evidence and validated with operational teams.

Communication: Summarize findings in a concise report for the management team, showing the relevant metric, comparison group, sample size, data limitation, and recommended follow-up.

Dataset

https://www.kaggle.com/datasets/suraj9727/supply-chain-optimization-for-a-fmcg-company

Data Dictionary

Column Name	Description
Ware_house_ID	Unique identifier for each warehouse
WH_Manager_ID	Identifier for the warehouse manager
Location_type	Type of location (Urban/Rural)
WH_capacity_size	Warehouse capacity category (Small/Mid/Large)
zone	Geographical zone of the warehouse
WH_regional_zone	Regional zone identifier
num_refill_req_l3m	Number of refill requests in the last 3 months
transport_issue_l1y	Number of transport issues in the last year
Competitor_in_mkt	Number of competitors in the market
retail_shop_num	Number of retail shops served by the warehouse
electric_supply	Availability of electric supply (1 = Yes, 0 = No)
dist_from_hub	Distance from central distribution hub (km)
workers_num	Number of workers in the warehouse
wh_est_year	Year of warehouse establishment
storage_issue_reported_l3m	Number of storage issues reported in the last 3 months
temp_reg_mach	Availability of temperature regulation machinery
approved_wh_govt_certificate	Type of government certificate approved
wh_breakdown_l3m	Number of warehouse breakdowns in the last 3 months
govt_check_l3m	Number of government checks in the last 3 months
product_wg_ton	Weight of products stored (tons)

Basic-Level Questions (Excel-Oriented)

1. What is the central tendency of warehouse capacity size?

- **Hint:** WH_capacity_size is a category (Small/Mid/Large), so use the mode or an ordered frequency summary. Do not calculate a numeric mean unless the categories have a defined numeric scale.
- **How it helps:** Shows which capacity category is most common and helps describe the warehouse network.
- **Business Impact:** Supports discussion of expansion or consolidation needs.

2. How many warehouses are located in urban areas versus rural areas?

- **Hint:** Count Ware_house_ID by Location_type using COUNTIF or a PivotTable.
- **How it helps:** Shows the distribution of warehouse locations and possible coverage differences.
- **Business Impact:** Supports network and accessibility planning.

3. What is the total number of retail shops served by each zone?

- **Hint:** Aggregate retail_shop_num by zone using a PivotTable.
- **How it helps:** Identifies which zones have higher retail shop coverage.
- **Business Impact:** Helps with resource allocation and inventory planning.

4. What is the average number of workers per warehouse?

- **Hint:** Use AVERAGE on workers_num, or summarize the average by zone for a more useful comparison.
- **How it helps:** Provides a benchmark for staffing requirements.
- **Business Impact:** Aids in labor-cost and productivity planning.

5. What percentage of warehouses have electric supply?

- **Hint:** Count warehouses with electric_supply = 1 and divide by the total number of warehouses.
- **How it helps:** Shows the share of facilities with recorded electric supply.
- **Business Impact:** Helps identify infrastructure upgrade needs.

6. What is the average distance of warehouses from the central distribution hub by zone and regional zone?

- **Hint:** Calculate average dist_from_hub by zone and WH_regional_zone in a PivotTable.
- **How it helps:** Assesses differences in warehouse location and potential logistics challenges.
- **Business Impact:** Informs transport planning and cost management.

7. How many warehouses reported storage issues in the last 3 months by zone and regional zone, and what percentage does this represent?

- **Hint:** Count warehouses with storage_issue_reported_l3m greater than zero by zone and WH_regional_zone; divide by warehouses in each group.
- **How it helps:** Highlights possible storage-management problems across groups.
- **Business Impact:** Guides improvement in storage practices and spoilage prevention.

8. Which three zones have the highest number of refill requests in the last 3 months?

- **Hint:** Aggregate num_refill_req_l3m by zone and sort descending to identify the top three.
- **How it helps:** Identifies areas with higher demand and potential supply-chain pressure.
- **Business Impact:** Helps improve inventory planning and stock availability.

9. What is the average number of government checks per zone and regional zone in the last 3 months?

- **Hint:** Calculate the average govt_check_l3m by zone and WH_regional_zone.
- **How it helps:** Shows how inspection activity varies across zones and regional zones.
- **Business Impact:** Assists compliance planning and regulatory-risk management.

10. What is the most common type of government certification among warehouses?

- **Hint:** Use the mode of approved_wh_govt_certificate or count each category in a PivotTable.
- **How it helps:** Identifies the most common recorded certification category.
- **Business Impact:** Helps maintain standards and compliance.

Medium-Level Questions (Excel-Oriented)

1. What is the correlation between the number of workers and the number of reported storage issues in the last 3 months?

- **Hint:** Compare workers_num with storage_issue_reported_l3m using a scatter plot and CORREL. Check outliers and avoid interpreting correlation as cause.
- **How it helps:** Shows whether staffing levels and reported storage issues move together.
- **Business Impact:** Helps identify facilities for staffing or process review.

2. How does warehouse capacity size relate to refill requests in the last 3 months?

- **Hint:** Use a PivotTable to compare num_refill_req_l3m across WH_capacity_size categories; include warehouse counts.
- **How it helps:** Compares refill activity among capacity groups.
- **Business Impact:** May inform replenishment planning by warehouse size.

3. Which geographical zone has the highest average number of transport issues in the last year?

- **Hint:** Group transport_issue_l1y by zone in a PivotTable and show both average and number of warehouses.
- **How it helps:** Locates where transport issues are most frequently reported.
- **Business Impact:** Helps prioritize route and logistics reviews.

4. What is the average product weight stored in warehouses with temperature regulation machinery?

- **Hint:** Compare product_wg_ton across temp_reg_mach categories using AVERAGEIF or a PivotTable. Verify how machinery availability is coded.
- **How it helps:** Summarizes stored product weight by temperature-control status.
- **Business Impact:** Supports review of equipment and storage requirements.

5. Which five warehouses had the highest number of government checks in the last 3 months, and what were their storage issue reports?

- **Hint:** Sort by govt_check_l3m, select the top five, and compare storage_issue_reported_l3m.
- **How it helps:** Checks whether frequently inspected facilities also report storage issues.
- **Business Impact:** Aids inspection preparation and storage follow-up.

6. How does the average number of workers compare between urban and rural warehouses?

- **Hint:** Use a PivotTable to compare workers_num by Location_type.
- **How it helps:** Identifies whether staffing levels vary by location type.
- **Business Impact:** Guides staffing policy discussions based on location requirements.

7. What is the relationship between distance from the hub and transport issues?

- **Hint:** Create a scatter plot of dist_from_hub and transport_issue_l1y; optionally calculate correlation.
- **How it helps:** Examines whether longer distances are associated with more reported transport issues.
- **Business Impact:** Can inform review of hub locations and transport routes.

8. How does competitor presence relate to refill requests?

- **Hint:** Compare Competitor_in_mkt with num_refill_req_l3m in a PivotTable; group competitor counts if needed.
- **How it helps:** Shows whether refill activity differs with local competitor presence.
- **Business Impact:** May inform inventory review in competitive markets.

9. Do storage issue reports differ between warehouses with and without government certificates?

- **Hint:** Compare storage_issue_reported_l3m across approved_wh_govt_certificate categories. Define the groups before any test; the certificate field may have multiple categories.
- **How it helps:** Assesses whether reported storage issues differ across certification categories.
- **Business Impact:** Guides further review of certification and storage practices.

10. How does warehouse establishment year relate to breakdowns in the last 3 months?

- **Hint:** Plot wh_est_year against wh_breakdown_l3m or compare grouped averages; account for missing years.
- **How it helps:** Identifies whether older and newer facilities report different breakdown counts.
- **Business Impact:** Informs maintenance and upgrade planning.

Advanced-Level Questions (Excel-Oriented)

1. Can transport issues be predicted from distance to hub and number of workers?

- **Hint:** Use transport_issue_l1y as the dependent variable and dist_from_hub and workers_num as predictors. Inspect residuals and limit conclusions to prediction.
- **How it helps:** Explores how distance and staffing relate jointly to reported transport issues.
- **Business Impact:** Can support exploratory planning and resource discussions.

2. Can warehouses be segmented by capacity, refill requests, and storage issues?

- **Hint:** Standardize numeric measures and encode WH_capacity_size before clustering; compare and validate cluster profiles.
- **How it helps:** Finds groups with similar observed characteristics.
- **Business Impact:** May help tailor operational reviews by facility segment.

3. How do operational measures vary by government certificate category?

- **Hint:** Compare wh_breakdown_l3m, storage_issue_reported_l3m, and govt_check_l3m by approved_wh_govt_certificate; show group sizes and account for other differences.
- **How it helps:** Summarizes how reported measures vary by certificate category without assuming certification caused the differences.
- **Business Impact:** Can guide further evaluation of certification and warehouse practices.

4. How can an Excel dashboard summarize warehouse performance?

- **Hint:** Use PivotTables, charts, and slicers for refill requests, transport issues, storage issues, and government checks per warehouse. Label each metric's time window.
- **How it helps:** Provides a concise view of warehouse measures and group comparisons.
- **Business Impact:** Improves visibility into operations and follow-up areas.

5. Can storage issues be forecast for the next 6 months?

- **Hint:** The listed fields provide counts over fixed retrospective windows, not dated monthly observations. Obtain dated historical counts before building a forecast.
- **How it helps:** Clarifies what additional data is needed for a valid time series.
- **Business Impact:** Prevents unsupported future values from being presented as reliable predictions.

Additional Considerations for Advanced Questions

- Data privacy and governance: Use warehouse and operational data only for authorized analysis; limit manager identifiers in shared outputs where they are not needed.
- Cross-functional validation: Review interpretations with warehouse managers, supply chain analysts, and operations teams.
- Continuous improvement: Refresh analysis when new records become available and evaluate whether operational changes correspond with measurable results.

Deliverables

- Case study document: problem statement, data dictionary, and analytical questions.
- Solution guide: detailed answers and explanations for each question.
- Additional resources: references for further exploration.

Desired Outcome

Trainees will develop an analytical and logical mindset, practice cleaning and summarizing operational data, and learn to communicate evidence-based findings. They will use Excel to explore warehouse capacity, replenishment, transport, storage, and inspection patterns and propose appropriate follow-up analysis.